郝宇

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我的研究集中在操作系统安全和 C/C++ 程序分析，近期工作包括将大语言模型用于安全分析和软件工程任务。我构建结合 Fuzzing、静态分析、符号执行和 LLM 推理的工具，用于 Linux、Android、鸿蒙内核等真实系统。相关工作包括 LLM 辅助静态分析、内核 Fuzzing 规格自动生成、漏洞可利用性分析和系统安全分析流程。研究成果发表在 S&P、USENIX Security、CCS、NDSS、ICSE、FSE、OOPSLA 和 TSE 等会议/期刊，并产出多个开源工具、Linux 内核漏洞/CVE 报告及补丁，同时受邀作报告。

教育经历

09/2018 – 09/2024 **博士**，加州大学河滨分校，计算机科学
导师：Zhiyun Qian

09/2015 – 06/2018 **硕士**，西安交通大学，控制科学与工程
导师：刘烃

09/2011 – 06/2015 **学士**，西安交通大学，控制科学与工程

09/2008 – 06/2011 呼和浩特第二中学，内蒙古自治区

工作经历

10/2024 – 至今 **系统安全研究员**
鸿蒙系统安全组，华为，中国
获得华为“天才少年”录用
鸿蒙系统安全、Fuzzing、LLM 辅助安全分析、静态分析

06/2023 – 09/2023 **研究实习生**，导师：Weidong Cui
系统安全与隐私研究组，微软研究院，美国
LLM 辅助安全分析与程序分析

06/2022 – 09/2022 **研究实习生**，导师：Kris Alder
Android 安全团队，谷歌，美国
Linux/Android 内核安全、Fuzzing、静态分析

09/2018 – 09/2024 **研究生研究员**，导师：Zhiyun Qian
加州大学河滨分校，网络安全与隐私研究和教育中心（CRESP）
操作系统安全、LLM 辅助安全分析、Fuzzing、符号执行、静态分析

06/2013 – 06/2018 **研究助理**，导师：刘烃
西安交通大学，智能网络与网络安全教育部重点实验室
并发安全、符号执行

毕业论文

[博士] Expanding the Boundaries of OS Kernel Fuzzing Permalink
- [C9, ICSE 22] [C10, S&P 23] [C17, CCS 25, Distinguished Paper Award]

[硕士] 基于并行程序不确定时序交织的恶意代码隐藏方法研究
- [C3, QRS 21, Best Paper Award] [J3, *IEEE Transactions on Reliability*]

出版论文

[C]: 会议, [J]: 期刊, [P]: 预印本
小计: CCF-A 类会议 12 篇 (一作 3 篇, 二作 6 篇), CCF-A 类期刊 1 篇, 正式发表论文 21 篇

[C18] NeuroMerge: ML-Guided State Merging for Efficient Symbolic Execution
ISSRE 26 Shenghan Zhng, Shitong Zhu, **Yu Hao**, Xingyu Li, Keyu Man, Zhang Zheng, Qing Deng, Zhiyun Qian, Srikanth Krishnamurthy
IEEE International Symposium on Software Reliability Engineering, ISSRE 26.

[C17] SyzSpec: Specification Generation for Linux Kernel Fuzzing via Under-Constrained Symbolic Execution
CCS 25 **Yu Hao**, Juefei Pu, Xingyu Li, Zhiyun Qian, Ardalan Amiri Sani
ACM SIGSAC Conference on Computer and Communications Security, CCS 25. Tool
- [Distinguished Paper Award]

[C16] S&P 25 SCAD: Towards a Universal and Automated Network Side-Channel Vulnerability Detection
Keyu Man, Zhongjie Wang, **Yu Hao**, Shenghan Zheng, Xin'an Zhou, Yue Cao, Zhiyun Qian
IEEE Symposium on Security and Privacy, S&P 25. Tool

[C15] USENIX Security 24 SymBisect: Accurate Bisection for Fuzzer-Exposed Vulnerabilities
Zheng Zhang, **Yu Hao**, Weiteng Chen, Xiaochen Zou, Xingyu Li, Haonan Li, Yizhuo Zhai, Zhiyun Qian, Billy Lau
USENIX Security Symposium 2024. Tool
- [Linux Security Summit 25]

[C14] OOPSLA 24 Enhancing Static Analysis for Practical Bug Detection: An LLM-Integrated Approach
Haonan Li, **Yu Hao**, Yizhuo Zhai, Zhiyun Qian
ACM SIGPLAN International Conference on Object-Oriented Programming Systems, Languages, and Applications, OOPSLA 24. Tool

[C13] S&P 24 SyzGen++: Dependency Inference for Augmenting Kernel Driver Fuzzing
Weiteng Chen, **Yu Hao**, Zheng Zhang, Xiaochen Zou, Dhilung Kirat, Shachee Mishra, Douglas Schales, Jiyong Jang, Zhiyun Qian
IEEE Symposium on Security and Privacy, S&P 24. Tool

[C12] NDSS 24 SyzBridge: Bridging the Gap in Exploitability Assessment of Linux Kernel Bugs in the Linux Ecosystem
Xiaochen Zou, **Yu Hao**, Zheng Zhang, Juefei Pu, Weiteng Chen, Zhiyun Qian
Network and Distributed System Security Symposium, NDSS 24. Tool

[P1] E&V: Prompting Large Language Models to Perform Static Analysis by Pseudo-code Execution and Verification
Yu Hao, Weiteng Chen, Ziqiao Zhou, Weidong Cui
- [AGI Leap Summit 2024] [Symposium on the Science of Security 24]

[C11] FSE 23 IVR Assisting Static Analysis with Large Language Models: A ChatGPT Experiment
Haonan Li, **Yu Hao**, Yizhuo Zhai, Zhiyun Qian
The ACM International Conference on the Foundations of Software Engineering, Ideas, Visions and Reflections, FSE 23 IVR. Tool

[C10] S&P 23 SyzDescribe: Principled, Automated, Static Generation of Syscall Descriptions for Kernel Drivers
Yu Hao, Guoren Li, Xiaochen Zou, Weiteng Chen, Shitong Zhu, Zhiyun Qian, Ardalan Amiri Sani
IEEE Symposium on Security and Privacy, S&P 23. Tool Result
- [Linux Security Summit 23] [Qualcomm Product Security Summit 23]
- [Symposium on the Science of Security 24]

[C9] ICSE 22 Demystifying the Dependency Challenge in Kernel Fuzzing
Yu Hao, Hang Zhang, Guoren Li, Xingyun Du, Zhiyun Qian, Ardalan Amiri Sani
IEEE/ACM International Conference on Software Engineering, ICSE 22. Tool
- [Google Research Paper Rewards]

[C8] NDSS 22 Progressive Scrutiny: Incremental Detection of UBI bugs in the Linux Kernel
Yizhuo Zhai, **Yu Hao**, Zheng Zhang, Weiteng Chen, Guoren Li, Zhiyun Qian, Chengyu Song, Manu Sridharan, Srikanth V. Krishnamurthy, Trent Jaeger, Paul Yu
Network and Distributed System Security Symposium, NDSS 22. Tool
- [2023 Cyber Security CRA Capstone Poster]

[C7] ACSAC 21 Eluding ML-based Adblockers With Actionable Adversarial Examples
Shitong Zhu, Zhongjie Wang, Xun Chen, Shasha Li, Keyu Man, Umar Iqbal, Zhiyun Qian, Kevin S Chan, Srikanth V Krishnamurthy, Zubair Shafiq, **Yu Hao**, Guoren Li, Zheng Zhang, Xiaochen Zou
Annual Computer Security Applications Conference, ACSAC 21. Tool

[C6] CCS 21 Themis: Ambiguity-Aware Network Intrusion Detection based on Symbolic Model Comparison
Zhongjie Wang, Shitong Zhu, Keyu Man, Pengxiong Zhu, **Yu Hao**, Zhiyun Qian, Srikanth V. Krishnamurthy, Tom La Porta, Michael J. De Lucia
ACM SIGSAC Conference on Computer and Communications Security, CCS 21. Tool

[C5] CCS 21 Statically Discovering High-Order Taint Style Vulnerabilities in OS Kernels
Hang Zhang, Weiteng Chen, **Yu Hao**, Guoren Li, Yizhuo Zhai, Xiaochen Zou, Zhiyun Qian
ACM SIGSAC Conference on Computer and Communications Security, CCS 21. Tool

[C4] FSE 20 UBITect: A Precise and Scalable Method to Detect Use-before-Initialization Bugs in Linux Kernel
Yizhuo Zhai, **Yu Hao**, Hang Zhang, Daimeng Wang, Chengyu Song, Zhiyun Qian, Mohsen Lesani, Srikanth V. Krishnamurthy, Paul Yu
ACM SIGSOFT International Symposium on Foundations of Software Engineering, FSE 20. Tool
- [2023 Cyber Security CRA Capstone Poster]

[J3] ConcSpectre: Be Aware of Forthcoming Malware Hidden in Concurrent Programs
Yang Liu, Ming Fan, Ting Liu, **Yu Hao**, Zisen Xu, Kai Chen, Hao Chen, and Yan Cai
IEEE Transactions on Reliability. Code Result

[C3] QRS 21 ConcSpectre: Be Aware of Forthcoming Malware Hidden in Concurrent Programs
Yang Liu, Ming Fan, Ting Liu, **Yu Hao**, Zisen Xu, Kai Chen, Hao Chen, and Yan Cai
IEEE International Conference on Software Quality, Reliability, and Security, QRS 21. Code Result
- [Best Paper Award]

[J2] TSE Tell You a Definite Answer: Whether Your Data is Tainted During Thread Scheduling
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, **Yu Hao**, Pei Liu, Ting Liu
*IEEE Transactions on Software Engineering, **TSE**.* Tool Benchmarks Result
- [S&P 17 Poster] Patent: [PCT] [CN]

[J1] Debugging Multithreaded Programs as if They Were Sequential
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, **Yu Hao**, Pei Liu, Lechen Yu, Ting Liu
IEEE Access. Tool

[C2] ICST 17 Automated Testing of Definition-Use Data Flow for Multithreaded Programs
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, Pei Liu, Jialiang Chang, **Yu Hao**, Ting Liu
IEEE International Conference on Software Testing, Verification and Validation, ICST 17. Tool

[C1] SATE 17 Debugging Multithreaded Programs as if They Were Sequential
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, **Yu Hao**, Pei Liu, Lechen Yu, Ming Fan, Ting Liu
IEEE International Conference on Software Analysis, Testing and Evolution, SATE 16. Tool

专利

WO2017181628 - TAINT ANALYSIS METHOD EMPLOYING SYMBOLIC COMPUTATION AND USED FOR DYNAMIC PARALLEL PROGRAM [PCT] [CN]

演讲，报告

- E&V: Prompting Large Language Models to Perform Static Analysis by Pseudo-code Execution and Verification
  - *AGI Leap Summit 2024* Website
  - *Symposium on the Science of Security, HoTSoS 2024* Website
- SyzDescribe: Principled, Automated, Static Generation of Syscall Descriptions for Kernel Drivers
  - *Linux Security Summit North America 2023* Website Slides Recording
  - *Qualcomm Product Security Summit 2023* Website
  - *44th IEEE Symposium on Security and Privacy, S&P 2023* Website Recording
  - *Symposium on the Science of Security, HoTSoS 2024* Website
- Demystifying the Dependency Challenge in Kernel Fuzzing
  - *44rd IEEE/ACM International Conference on Software Engineering, ICSE 2022* Website Slides Recording

荣誉，奖励

2025 CCS 2025, Distinguished Paper Award

2024 Laxmi Bhuyan Fellowship Award, University of California, Riverside

Dissertation Completion Fellowship Award, University of California, Riverside

2023 IEEE S&P Student Travel Grant

Google Research Paper Rewards

The Linux Foundation's Travel Fund

2021 QRS 2021 Best Paper Award

2018 Dean's Distinguished Fellowship, University of California, Riverside

西安交通大学优秀硕士毕业生

2013 西安交通大学程序设计竞赛，二等奖

西安交通大学数学建模竞赛，二等奖

2010 内蒙古高中数学联赛，一等奖

内蒙古高中物理联赛，三等奖

内蒙古高中化学联赛，三等奖

2007 全国奥林匹克数学竞赛，金牌

学术服务

程序委员会委员	CCS 2027, 2026. RAID 2026. FORGE@ICSE 2025, 2024. MSR 2024(Junior PC)
审稿人	TIFS 2026, 2024, 2023. TDSC 2026. TOSEM 2026, 2025. Computing Surveys 2026, EAI SecureComm 2023
外部审稿人	S&P 2025, 2024, 2021, 2020. CCS 2024. USENIX Security 2025, 2021. NDSS 2025, 2021, 2020
AEC	EuroSys 2023. ISSTA 2024. ECOOP 2024

推荐人

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